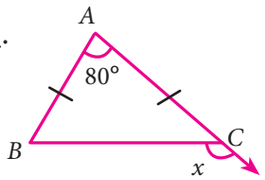
20. An exterior angle of a triangle is 70° and two interior opposite angles are equal. Then measure of each of these angle will be

(i) 110° (ii) 120° (iii) 35° (iv) 60°

21. In a ΔABC, $AB = AC$. The value of x is _____.

(i) 80° (ii) 100°

(iii) 130° (iv) 120°



22. If an exterior angle of a triangle is 115° and one of the interior opposite angles is 35°, then the other two angles of the triangle are

(i) 45°, 60° (ii) 65°, 80° (iii) 65°, 70° (iv) 115°, 60°

4.4 Congruency of Triangles

We are now going to learn a very important geometrical concept called congruence. To understand the concept of congruency of triangles, let us first look into the congruency of shapes.

4.4.1 Congruency of Shapes

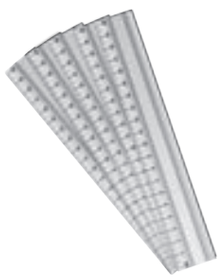
Let us observe the following pictures.



Currency notes of same denomination



A pack of playing cards



Rulers of same size

Fig. 4.17

To understand congruency of shapes, let us take a pack of cards and choose any two cards and place them one over the other. We are able to place them in such a way that one card matches exactly with the other in size and shape. Hence all the cards in a pack are equal in both size and shape.

Any pair or set of objects with this property are said to be congruent.

How can we check the congruence of two objects or figures?

To check the congruence of the objects, we can use the method of superposition. In this method we have to make a trace-copy of one figure and place it over the other. If the figures match each other completely then, they are said to be congruent. In this method to match the original with the trace copy, we are not allowed to bend, twist or stretch, but we can translate or rotate.

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